

Know your supplier is going to be late — before they do.

A live digital twin of an SME's supply chain that predicts a disruption at least 48 hours out, names an alternative supplier, and hands the owner a plan they can act on in minutes.

CHALLENGE

du SME Resilience & Innovation

TRACK

Theme 1 · ResilienceTech

THEME

AI-Driven Supply Chain Digital Twins

STATUS

Working system, not a concept

The information existed. Nobody had time to assemble it.

TODAY · DELIVERY MORNING

The van doesn't arrive.

Both remaining options are bad: pay a premium for an emergency order, or tell customers the shelf is empty.

WITH SUPPLYTWIN · 48 HOURS EARLIER

"Gulf Wholesale is likely to slip. Desert Star can cover it."

Named alternative, step-by-step plan, owner decides.

The signals were already there — this supplier slipped three times in thirty days, and that item is below its reorder level. Larger firms catch this with supply-chain planners and six-figure ERP modules. An SME has neither the headcount nor the licence.

One live model of stock, suppliers and orders — that hands over a decision, not a dashboard.

Ingest

CSV, manual entry, or API. Setup takes about two minutes and needs no IT project — because an SME that needs an integration project will never start.

Predict

Supplier signals are evaluated against the twin. Every alert carries a confidence score and a rationale written in the owner's language.

Act

An alternative supplier and a numbered plan. The owner accepts, adjusts or dismisses. Nothing happens without them.

Track: **Resilience** — Theme 1, AI-Driven Supply Chain Digital Twins.

Six steps, and the owner holds the last one.

01 Ingest Stock, suppliers, open orders — however the business already keeps them.	02 Model A live twin: lead times, reorder levels, which supplier serves which item.	03 Predict A disruption prediction with confidence and a plain-language rationale.
04 Source The business's <i>own</i> backup supplier first, then verified local suppliers.	05 Plan Numbered steps in the language an owner uses, not a planner's.	06 Decide Accept, adjust, dismiss. Automatic action is opt-in, per supplier.

Three decisions we would defend in front of a regulator.

The AI doesn't set the guarantees

The 48-hour floor and the sourcing priority are ordinary code with tests around them. A model that drifts changes the wording of an alert — it cannot change how early the alert fires.

The database enforces isolation

PostgreSQL row-level security, forced, under a non-superuser role. Missing tenant context returns zero rows, never every row. It fails closed.

It shows what it got wrong

Every warning is listed, including the ones that came to nothing. An owner who cannot see the misses has no basis for trusting the hits.

At 40,000 SMEs on shared infrastructure, an application-layer tenant check is one forgotten WHERE clause away from a breach. That is why this one lives in the database.

API-first, contract-first — and du already has the relationship.

Three ingestion paths, by SME maturity

- **CSV upload** — businesses running on spreadsheets, which is most of them
- **Manual entry** — the smallest, with a handful of SKUs
- **REST + webhooks** — anyone with a POS or accounting system

The OpenAPI 3 contract is the source of truth, lives in the repository and is served at /docs. A test parses it on every commit.

Where it meets du

- **Identity** — du SSO replaces the onboarding module without touching the twin
- **Distribution** — the pilot cohort is a segment query, not a campaign
- **Notification** — du's SMS or WhatsApp gateway is a provider, not a redesign
- **Directory** — a du-verified supplier list drops in as a data source

Three containerised services over shared PostgreSQL and Redis. Nothing is tied to one cloud, and the language model sits behind an OpenAI-compatible client — the provider is three environment variables, already exercised against more than one vendor.

The cohort widens only when the last stage has earned it.

5 Design partners · weeks 1–2 Food distribution, retail and spare parts. The goal is not usage metrics — it is having a practitioner correct the system's judgement. EXIT TEST Three of five confirm the plans read like something they would actually send.	25 Closed pilot · weeks 3–6 A du segment query supplies the cohort: SMEs holding physical stock with more than one supplier. Onboarding is self-serve; we watch where people stop. EXIT TEST ≥80% onboard unaided · ≥1 real disruption caught ≥48h out · zero isolation incidents.	150+ Scale pilot · weeks 7–16 Three waves of roughly fifty, so a defect found in wave one never reaches wave three. Automatic action stays opt-in throughout. EXIT TEST The success metrics, measured against each business's own first four weeks.
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A system that mispredicts for 150 businesses at once does not get a second chance with them.

Targets the pilot is designed to validate — not results already claimed.

KPI	BASELINE	TARGET	MEASURED BY
Warning lead time	0h	≥ 48h	Alert timestamp vs predicted impact date. Enforced in code
Stockout reduction	Own weeks 1-4	≥ 40%	Items below reorder with no inbound order, before vs after
Warning → action	Days	< 10 min	Alert delivered → owner decision recorded
Alternative found	—	≥ 90%	Alerts carrying a named alternative ÷ all alerts
Owner acceptance	—	≥ 50%	Plans accepted or adjusted rather than dismissed
False positive rate	—	≤ 30%	Published whatever it is — shown in-product, not just to du
Onboarding completion	—	≥ 80%	Signup → first twin view, unaided, median under 5 min

"Warning → action" is the brief's *business pivot speed*: moving an order to a different supplier goes from days to minutes.

The risks that would actually end a pilot.

RISK	MITIGATION
Cross-tenant data leak One leak ends the pilot	Enforced in PostgreSQL, not the app — RLS forced, non-superuser runtime role, fails closed on missing context, covered by isolation tests
A wrong recommendation costs real money	The owner decides. Automatic action is off by default, opt-in per supplier, gated on a confidence threshold they set. Every recommendation is audit-logged with its inputs
Model drift or a provider change	The guarantees are not in the model. The 48-hour floor and sourcing priority are code with tests; the model writes prose and a score
PDPL and data residency	No data reaches a model without explicit, revocable consent — enforced as a 403, not a checkbox. Deployment is region-portable
Alert fatigue The classic failure of this category	False positive rate is a published KPI shown to the SME. Escalation is staged and the track record is visible, so trust is earned rather than assumed
Stale data served as current	A freshness checker marks sources stale and flags the twin rather than quietly serving last month's numbers

What is built, what is simulated, and what is not built yet.

BUILT & WORKING

The twin. Three agents against a live model. Consent enforcement. Row-level tenant isolation. Audit trail. Opt-in automatic action. The track-record page. The full API contract.

96 automated tests — 74 platform, 22 AI service.

SIMULATED

Supplier disruption signals. The prediction logic is real; the feed is seeded. Live carrier and port data is a pilot-stage integration, and is scoped as one rather than implied.

NOT BUILT YET

Real authentication — there is no password anywhere; the hosted demo signs in to one seeded account. Outbound email and SMS: intent is recorded, nothing is sent. The periodic sweep is written and tested but not yet deployed, so the hosted alert was raised by running the same pipeline against it once.

A load test at pilot scale — 150 tenants, 3,000 inventory items — returned alert delivery p95 of **234ms** and twin reads p95 of **50ms** against a five-minute budget. That is sequential latency on a developer laptop, not a concurrency measurement, and the report says so. Re-running it on the deployed environment is a gate before the 150-SME cohort, not after.

All three, and how.

Built for rapid scalability

Three independently scalable stateless services, containerised, over shared PostgreSQL and Redis. No per-customer infrastructure.

100% digital integration

Signup, data connection, consent, prediction, recommendation, decision and audit are all in-product. No manual or offline step in the core loop.

Scales to 40,000+ SMEs

Row-level isolation on shared tables rather than a database per customer. Validated at 150 tenants; the next gate is 1,500.

SUPPLYTWIN

Most SMEs find out on delivery day. This is the version where they find out first.

Built spec-first — constitution, spec, plan, tasks — with the full build log, the API contract and every test in the repository.

TRY IT

supply-chain-digital-twin-web.vercel.app

API

supply-chain-digital-twin-production.up.railway.app/docs

CODE

github.com/my5757980/supply-chain-digital-twin

SUBMITTED BY

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The live link opens a seeded demonstration account — one click, no sign-up. Everything on it is produced by the running system.